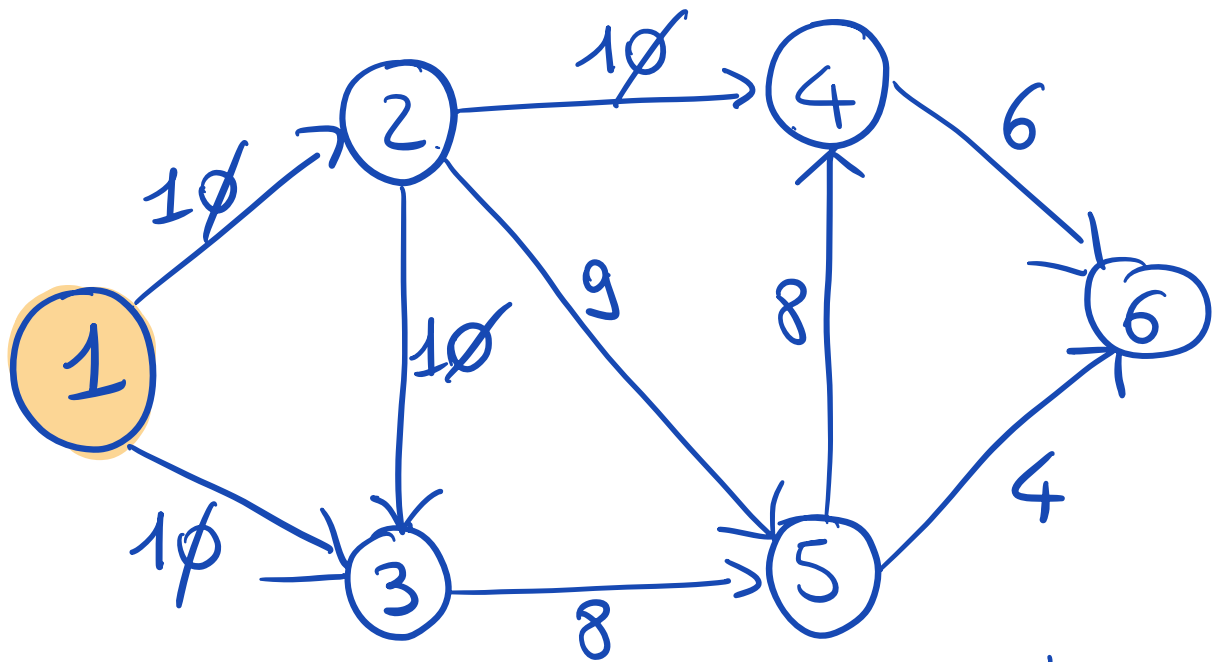
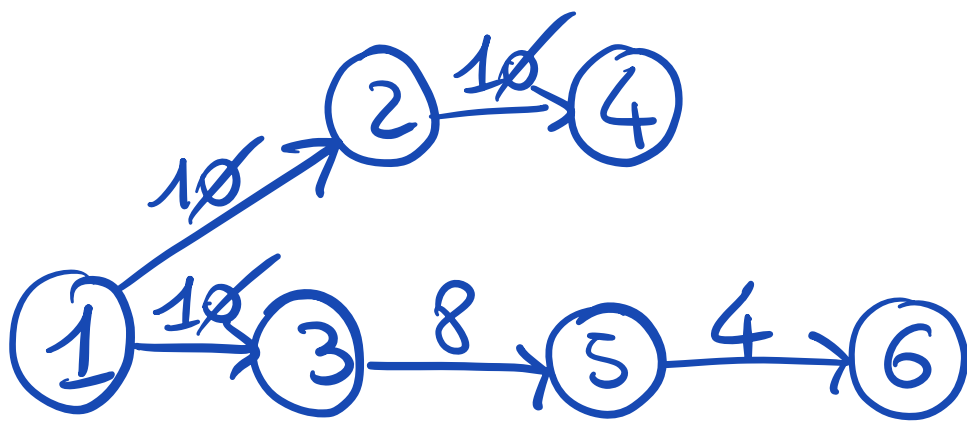




1	2	3	4	5	6
∞ NIL	∞ N	∞ N	∞ N	∞ N	∞ N
1	1 1	4 1	∞ N	∞ N	∞ N
2	2	1 1	2 2	19 2	∞ N
3	3	3	2 2	18 3	∞ N
4	4	4	2 2	4	22 5
5	5	5	5	5	22 5
6	6	6	6	6	6



$$T(n) = 3T(n/4) + \underbrace{n \cdot \log n}_{\text{METODO ESPERTO}}$$

$$1) \quad n^{\log a} = n^{\log 3} \approx n^{0.7}$$

$$2) \quad \text{confrontare } \underbrace{n^1 \cdot \log n}_{\text{METODO ESPERTO}} \text{ con } \underbrace{n^{\log 3}}_{\approx n^{0.7}}$$

CASO 3

$$n \cdot \log n = \Omega(n^{\log 3 + \epsilon})$$

$\epsilon < 0.3 \quad 0.1$

$$\boxed{\frac{3}{4} n \cdot \log\left(\frac{n}{4}\right) \leq c n \cdot \log n}$$

$c = \frac{3}{4}$

$$\frac{3}{4} \log \left(\frac{n}{4} \right) \leq \frac{3}{4} \log \underline{(n)} \quad \Leftarrow$$

$$\underline{\underline{T(n) = \Theta(n \cdot \log n)}} \quad \Leftarrow$$

....

```

for i = 1 to n - O(m) → J ≤ m²
{
  J = i + 1
  while (J/m ≤ m) - O(m²)
  {
    k = 1
    while (k ≤ m) - O(m)
    {
      P(4) /* p è di costo costante */
      k = k + 3
    }
    J = J + 1
  }
}

```

$O(m^4)$

$$\underline{T(n) = 2T(n) + \sqrt{n} + 42 \quad \leftarrow \text{NO}}$$

$$T(n) = 2T(n/2) + \sqrt{n} + 42$$

$$1) n^{\log_a b} = n^{\log_2 2} = n$$

$$2) \text{ confronto } \sqrt{n} + 42 \text{ con } n$$

$$\underline{n^{0.5} + 42} \quad \underline{n^1}$$

CASO 1

$$\sqrt{n} + 42 = O(n^{1-\epsilon})$$

$$\epsilon = 0.1$$

$$T(n) = \Theta(n)$$

$$T(n) = \begin{cases} \underline{3T(n/4) + cm^3} & \text{se } n > 4 \\ \underline{c} & \text{altrimenti:} \end{cases}$$

$$\frac{n}{4^i} = 4$$

$$n = 4^{i+1}$$

$$i+1 = \log n$$

$$i = \log n - 1$$

$$\sum_{i=0}^{\log n - 1} \left(\frac{3}{64}\right)^i \cdot c \cdot n^3$$



~~$\Theta(\log n)$~~

$\text{Algo}(A, i, j)$
 $\{$ if $A[i] > A[j]$
 then $\text{scambia}(A[i], A[j])$

if $(i+1 \geq j)$
 then return true

$$K = \lfloor (j-i+1)/3 \rfloor$$

$\text{Algo}(A, i, j-K)$

$\text{Algo}(A, i+K, j)$

$\text{Algo}(A, i, j-K)$

}

$$T(n) = 3 \cdot T\left(\frac{2}{3}n\right) + 1 + c + \Theta(1)$$

$$\underbrace{n^{\log_{3/2} 3}}_{\text{com } 1 = n^0}$$

CASO 1

$$n^0 = O\left(n^{\log_{3/2} 3 - \epsilon}\right)$$

$$\boxed{\Theta\left(n^{\log_{3/2} 3}\right)}$$

...
count = 0

J = 0

for $i = 0$ to $\log n \leftarrow O(\log n)$

$$\sum_{i=0}^{\log n} (n \cdot i) =$$

J = 2

while $J < 2^n \leftarrow O(m)$

$O(i) \rightarrow \text{count} = \text{count} + f(i)$

$J = 2 \cdot J$

1) istruzione = count++ $O(m \cdot \log n)$

2) istruzione = count = count + f(i)
 $f(m)$ ha complessità $O(m)$



$$1 + 2 + 3 + 4 + \dots \log n$$

$$n \cdot \frac{\log n \cdot (\log n + 1)}{2}$$

2

$$O(m \cdot (\log n)^2)$$

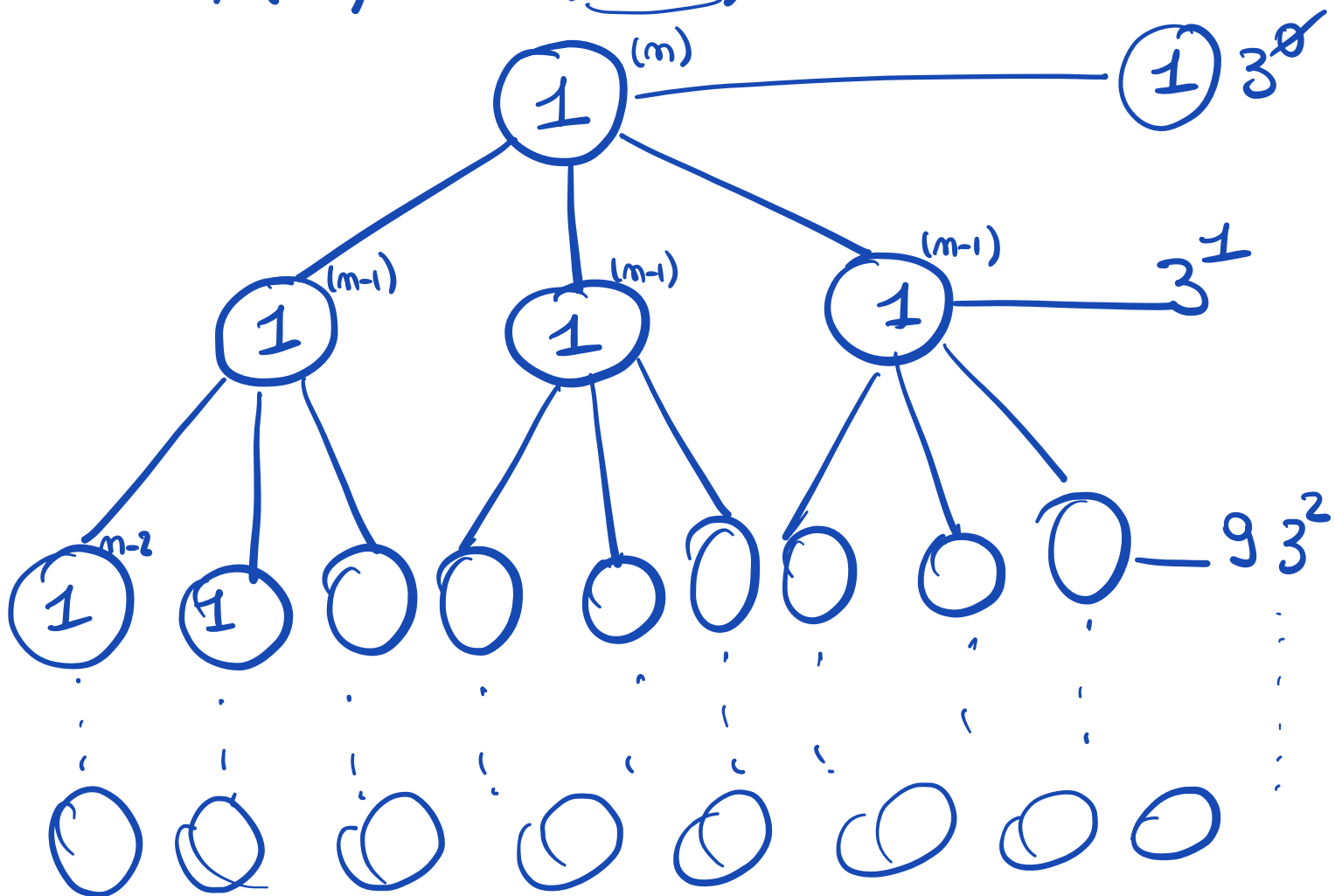
$f_1(n)$

{ if $n \leq 1$ then return 1

else

} return $\underbrace{f_1(n-1)} + \underbrace{f_1(n-1)} + \underbrace{f_1(n-1)}$

$$T(n) = 3T(n-1) + \Theta(1)$$



$$3^0 + 3^1 + 3^2 + \dots + 3^m \Rightarrow$$

$$O(3^m)$$